

Pack Out a Finished Unit SOP (Foam, Unit, Manual, Top Pad, Flaps, H-Tape)

This SOP covers packing one finished unit into a shipping carton using a two-arm robot system. Both arms are used throughout: the grippers cooperate to seat the bottom foam insert, lower the unit into its cavity, add the manual, lay the top foam pad over the load, fold the carton flaps closed and lay an H-pattern of pre-cut tape strips across the top. The task runs from an open empty carton at the center and a finished unit in its start zone to one sealed, taped carton at the carton station, with the table clear at the end.

The table is set up in one of three ways. Only the unit moves; the carton, the foam source, the manual station and the tape station are in the same place in all three.

- **Config L:** the unit is at the front-left, in front of the foam source.
- **Config M:** the unit is at the back-center, behind the carton.
- **Config R:** the unit is at the front-right, in front of the tape station.

Where a step depends on the setup it says so on an **IF** line — look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the unit starts at the front-left (**Config L**), the back-center (**Config M**) or the front-right (**Config R**). One config per episode, chosen before recording and never changed mid-episode.
- **Same-side rule:** the gripper on the unit's side places it — the left gripper in Config L and M, the right gripper in Config R. No arm reaches across the table.
- **Fixed roles:** in every config the left gripper seats the bottom foam insert and lays the top foam pad from the left-side foam source, the right gripper adds the manual, and both grippers land every tape strip. Only the unit pick changes hands.
- **The carton starts open:** the carton is already at the center, sitting flat and square, with all four flaps standing upright. It is not squared, opened or set up on camera.
- **One carton at a time:** exactly one carton is loaded, closed and taped before the next carton is started.
- **Pack order is fixed:** the carton is loaded bottom-up in one order only: **bottom foam insert, unit, manual, top foam pad**. The order is never changed and no layer is skipped.
- **The carton stays down:** the carton sits flat on the table at the center for the whole pack-out. It is never lifted, tilted or turned over while it is open.

- **Everything is lowered, never dropped:** the unit and the manual are carried into the carton and lowered until they touch down before the gripper opens.
- **Flaps close side flaps first:** the two short (minor) flaps on the left and right ends fold in first, then the two long (major) flaps at the near and far sides fold down over them.
- **Tape is pre-cut and two-handed:** every strip is a pre-cut piece taken from the tape station. The right gripper takes one side of the strip, the left gripper takes the other, and both grippers work together to align and land it. There is no tape gun.
- **H-pattern tape:** the top is always sealed with three strips, the center seam strip first, then the right end strip, then the left end strip, forming an H.
- **Nothing loose:** the carton is packed so that nothing shifts or rattles inside when it is pressed.

Setup

Go through both checklists before starting the episode.

Hardware checklist

- Cameras are on and recording
- Env camera frame includes the front and back edges of the table and is centered on the table's midpoint
- Env camera frame includes the open carton mouth, all four flaps, the tape station and the unit's start zone
- Both arms are at the home position with grippers open
- Table surface is clear of any objects other than the stations listed below

Materials checklist

- One finished unit is in the **start zone** for this episode's config, in its sealed bag:
 - **Config L:** front-left, in front of the foam source
 - **Config M:** back-center, behind the carton
 - **Config R:** front-right, in front of the tape station
- One empty carton is at the **center**, sitting flat, square to the table edge, with all four top flaps open and standing upright
- The carton's bottom is already sealed and holds its shape when pressed
- Foam is at the **left side**: bottom foam inserts cavity side up and top foam pads, laid side by side and not stacked, both matching the unit model
- Manuals are at the **front-center** manual station, in one flat pile

- Three **pre-cut tape strips** are staged at the **center-right** tape station, laid out flat and tacked by one edge so a gripper can lift each one clear

Workspace layout

- **Start zone:** unit staging (the finished unit, input) — front-left (Config L), back-center (Config M) or front-right (Config R)
- **Center:** carton station (the open carton being packed, and the sealed carton at the end). It stays in the center and is never moved.
- **Left side:** foam source (bottom foam inserts cavity side up and top foam pads, laid side by side, not stacked)
- **Front-center:** manual station (the pile of manuals)
- **Center-right:** tape station (the three pre-cut tape strips)

Vocabulary

These are the terms used in this SOP. Operators and annotators must use this language consistently. One term per concept, used throughout.

- **Carton:** the shipping box the unit is packed into. It sits at the center for the whole episode.
- **Carton mouth:** the open top of the carton, the target everything is lowered through.
- **Carton floor:** the inside bottom face the bottom foam insert sits on.
- **Carton wall:** one of the four inside faces of the carton.
- **Minor flap:** one of the two short top flaps, on the left and right ends of the carton. These are the side flaps and they fold in first.
- **Major flap:** one of the two long top flaps, at the near and far sides of the carton. These fold down over the minor flaps.
- **Center seam:** the line down the middle of the closed top where the two major flap edges meet.
- **End seam:** the line across each end of the closed top where a major flap edge meets the carton wall.
- **Flap gap:** an opening left along the center seam because a flap is not folded all the way in or the flaps overlap.
- **Bottom foam insert:** the shaped foam tray that sits on the carton floor and holds the unit.
- **Cavity:** the recess in the bottom foam insert that the unit body drops into.
- **Keyed corner:** the cut corner of the bottom foam insert that matches one corner of the carton and sets its orientation.
- **Top foam pad:** the flat foam pad laid over the load, under the flaps, as the last layer in the carton.
- **Unit:** the one finished product being packed, in its sealed bag.

- **Start zone:** where the unit lies at the start of the episode — front-left (**Config L**), back-center (**Config M**) or front-right (**Config R**). One per episode, chosen before recording and never changed mid-episode.
- **Manual:** the printed booklet that ships with the unit, laid flat on top of it in the carton.
- **Void:** an empty gap inside a packed carton that lets contents shift.
- **Tape strip:** one pre-cut length of tape from the tape station. Three strips seal one carton.
- **Center strip:** the strip landed along the center seam, running from one end of the carton to the other.
- **End strip:** a strip landed across an end seam, at a right angle to the center strip.
- **Wing:** the part of a strip that runs down a carton wall past the top edge.
- **H-pattern:** the finished pattern of one center strip crossed by two end strips.
- **Seated:** a bottom foam insert lying flat on the carton floor, corners down, with no lift at any edge.
- **Nested:** a unit lowered into the cavity so its base sits on the cavity floor and the foam holds it on all sides.
- **Loaded:** a carton holding the bottom foam insert, the unit, the manual and the top foam pad, in that order.
- **Closed:** a carton whose minor flaps are folded in and whose major flaps lie flat over them.
- **Sealed:** a closed carton with the full H-pattern landed and lying flat.
- **Carton station:** the center position where the carton sits for the whole pack-out and stays at the end. It is never moved.
- **Foam source:** the left side of the table where the bottom foam inserts and top foam pads lie side by side, not stacked.
- **Manual station:** the front-center pile of manuals.
- **Tape station:** the center-right area holding the three pre-cut tape strips.
- **Home position:** the default resting pose for each arm: gripper open and clear of the table.
- **Seat:** lower the bottom foam insert onto the carton floor and press it flat.
- **Place:** lower the unit into the cavity until it touches down, then release.
- **Add:** lower the manual into the carton on top of the unit and foam.
- **Fold:** bring a flap down over the carton mouth.
- **Land:** bring a pre-cut tape strip down onto a seam with both grippers and let it take hold.
- **Press:** an optional adjustment: run either gripper along a landed strip so that it applies well.
- **Sweep:** check every station is clear at the end of the episode.

Steps

Small corrections are expected and allowed. Settling a layer, a flap or a strip with a few adjustments is normal work, and so is opening and re-gripping when a manual, a foam pad or a

tape strip sticks to a gripper. Neither is a violation.

Only Step 2 depends on where the unit starts: in Config L and M the **left gripper** takes the unit from the start zone; in Config R the **right gripper** takes it. Every other line is the same in all three configs.

Step 1: Seat the bottom foam insert

Goal: one bottom foam insert lies flat on the carton floor, cavity side up, keyed corner matched.

- With the left gripper, pinch a bottom foam insert (the foam pad with the cavity in it) by its **near edge** and lift it off the left-side foam source. The near edge is the only pick point; the insert is never taken by a far edge, a side edge or the cavity.
- Carry it over the carton mouth, cavity side up, and turn it until its **keyed corner** lines up with the matching corner of the carton.
- Lower it straight down between the walls until it touches the carton floor, then open the gripper.
- Press the insert down with a gripper until the whole insert lies flat with no lift at any edge.
- If the insert will not lie flat, or one edge rides up a carton wall, lift it clear, re-aim on the keyed corner and lower it again.
- Do not lift or tilt the carton at any point from here until it is sealed.

Step 2: Place the unit in the cavity

Goal: the unit sits down in the foam cavity, held on all sides.

Look where the unit is before reaching for it.

- **IF the unit is at the front-left (Config L):** with the **left gripper**, grasp the unit at the middle of its body and carry it right, over the carton mouth.
- **IF the unit is at the back-center (Config M):** with the **left gripper**, grasp the unit at the middle of its body and carry it forward, over the carton mouth.
- **IF the unit is at the front-right (Config R):** with the **right gripper**, grasp the unit at the middle of its body and carry it left, over the carton mouth.

Then, in all three:

- Line the unit's body up with the cavity below it.
- Lower it straight down into the cavity, keeping the body square to the walls, until the base touches the cavity floor, then open the gripper.
- **Lower, never drop:** the unit is released only after it has touched down inside the cavity.
- Confirm the unit sits down in the cavity on all sides, with no corner riding on the foam. If it is not well seated, adjust it with the gripper that placed it.

Step 3: Add the manual

Goal: the manual lies flat on top of the unit and foam, inside the carton walls.

- With the right gripper, take one manual from the front-center manual station.
- Carry the manual over the carton mouth and lower it flat across the top of the unit and foam, printed side up and inside the carton walls, then release.
- If the manual hangs over a carton wall or blocks a flap from folding, push it down inside the walls with the right gripper.
- Press down once across the load with the right gripper. If anything shifts or a **void** is left along a wall, re-seat that layer before the top pad goes in.

Step 4: Lay the top foam pad over the load

Goal: one top foam pad lies flat over the load, inside the carton walls, below the top edge.

- With the left gripper, grab a top foam pad at the **middle of its near edge**, lift it off the left-side foam source and carry it over the carton. The middle of the near edge is the only pick point; the pad is never taken by a corner or a side edge.
- Lower it flat onto the manual and the load, keeping it square to the walls, then open the gripper.
- With the left gripper, press the pad down until it lies flat, fully inside the carton walls, with no corner riding up a wall and nothing above the top edge.
- If the pad stands above the top edge, lift it clear, push the load down inside the walls and lay it again. Never close a flap over a raised pad.

Step 5: Close the carton flaps

Goal: the four flaps lie flat and closed, side flaps under major flaps, ready for tape.

- With the left gripper, fold the left **minor flap** (the side flap) in flat over the top foam pad.
- With the right gripper, fold the right **minor flap** in flat over the top foam pad.
- Both side flaps go in before either major flap is touched.
- With the left gripper, fold the far **major flap** down over the minor flaps.
- With the right gripper, fold the near **major flap** down until its edge meets the far flap edge.
- Confirm the two major flap edges meet along the **center seam**, with neither flap riding over the other.
- If a flap will not lie flat, or a flap edge stands up off the seam, open the flaps, push the load down inside the walls and close again. Never force a flap down over a raised load.

Step 6: Tape the H-pattern

Goal: the top is sealed with three pre-cut strips, one along the center seam and one across each end seam, forming an H.

Every strip is handled the same way: the right gripper lifts the strip by its right side, the left gripper takes the left side, both grippers line the strip up with the seam, and then both grippers work together to land it. Pressing a landed strip is an optional adjustment that either gripper may make.

6.1 Land the center strip

- With the right gripper, take a pre-cut tape strip from the center-right tape station by its right side.
- With the left gripper, take the same strip by its left side, so both grippers hold it with the strip stretched out between them.
- With both grippers, line the strip up with the **center seam**, running from one end of the carton to the other.
- With both grippers working together, land the strip on the center seam, keeping it centered on the seam along its whole length, and carry each end over the top edge and down the wall by about the same amount, so both **wings** lie flat.
- If the strip needs settling, either gripper may press along it, end to end, including both wings, so that it applies well. This press is an adjustment, not a required step, and skipping it is not a violation.

6.2 Land the right end strip

- With the right gripper, take the second pre-cut strip from the tape station by its right side.
- With the left gripper, take the same strip by its left side.
- With both grippers, line the strip up with the right **end seam**, at a right angle to the center strip.
- With both grippers working together, land the strip across the right end seam, carry each end over the top edge and down the side wall by about the same amount, so that it lies flat. Either gripper may be used to press the strip down so that it applies well.

6.3 Land the left end strip

- With the right gripper, take the third pre-cut strip from the tape station by its right side.
- With the left gripper, take the same strip by its left side.
- With both grippers, line the strip up with the left **end seam**, parallel to the right end strip.
- With both grippers working together, land the strip across the left end seam, carry each end over the top edge and down the side wall by about the same amount, so that it lies flat. Either gripper may be used to press the strip down so that it applies well.
- Confirm the finished pattern is an **H**: one strip along the center seam crossed by one strip at each end.
- Do not land any extra strips. Three strips seal the carton.

Step 7: Confirm the pack-out and end the episode

Goal: the sealed carton stands square at the carton station and the stations are ready for the next unit.

- Confirm the center seam and both end seams are covered by the three strips.
- Confirm the sealed carton sits flat and square at the carton station.
- Confirm the table holds no foam, manual or tape offcut.
- Move both arms back to the home position with grippers open.
- End data collection.

SOP violations

These are the things the operator can do that break (violate) the SOP. This list feeds the violation set and is what reviewers look for using the side-by-side review tool.

How to record a violation in review

For each violation you spot in a recorded episode, record:

- The **start timestamp** of the violation in the video.
- The **violation name** from the list below.
- The **SOP rule broken** (the step number from the list below).

The **visible cue** is what you actually see in the video. The **coaching note** is for retraining the operator after the review; it is not what the annotator labels.

Episode handling

Any SOP violation is flagged (or tagged) with the timestamp and violation name. Rather than being discarded, the episode is retained in the training data and tagged with the violation. No flagged episode is deleted. This matches the goal of capturing realistic operator variance in training.

Violations

Note on the start position: the violations below were written for Config R (the unit at the front-right, taken by the right gripper). The pickup and arm-role cues will be rewritten later to cover all three start positions; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned.

- **Visible cue:** what the operator does does not match the config on the table — the unit is not in the start zone for the config; a gripper reaches across the table for the unit; or the wrong IF line is followed.

- **SOP rule broken:** the start position and the same-side rule (the left gripper takes the unit from the start zone in Config L and M, the right gripper in Config R; no arm reaches across the table; the IF line followed is the one for the config on the table).
- **Coaching note:** look where the unit is before the first reach, then follow that config's IF line through Step 2.

Violation: Pack order broken.

- **Visible cue:** a layer goes in out of order: the unit is placed before the bottom foam insert is seated, the manual goes in before the unit, or the top foam pad goes in before the manual.
- **SOP rule broken:** Steps 1 to 4 (load the carton bottom-up: bottom foam insert, unit, manual, top foam pad).
- **Coaching note:** load in one fixed order every time. Bottom foam first, then the unit, then the manual, then the top pad.

Violation: Bottom foam insert skipped.

- **Visible cue:** the unit is lowered straight onto the carton floor with no foam insert under it, or the insert is left at the foam source while the carton is closed.
- **SOP rule broken:** Step 1 (seat one bottom foam insert on the carton floor before the unit goes in).
- **Coaching note:** the insert is what holds the unit off the carton floor. No unit goes in on bare board.

Violation: Bottom foam insert seated wrong way up or wrong way round.

- **Visible cue:** the insert goes in cavity side down, or its keyed corner does not line up with the matching carton corner, so the cavity sits across the wrong axis.
- **SOP rule broken:** Step 1 (cavity side up, keyed corner matched to the carton corner).
- **Coaching note:** read the cavity and the keyed corner before lowering. A turned insert leaves the unit unsupported on two sides.

Violation: Bottom foam insert not lying flat.

- **Visible cue:** an edge or corner of the insert rides up a carton wall, or the insert is domed off the carton floor, and the operator carries on loading.
- **SOP rule broken:** Step 1 (press the insert down until it lies flat with no lift at any edge).
- **Coaching note:** press all four corners before the unit goes in. An insert that rides up puts the unit in crooked.

Violation: Foam picked by the wrong point.

- **Visible cue:** the bottom foam insert is taken by a far edge, a side edge or the cavity instead of pinched by its near edge, or the top foam pad is taken by a corner or a side edge instead of the middle of its near edge.

- **SOP rule broken:** Steps 1 and 4 (pinch the bottom foam insert by its near edge; grab the top foam pad at the middle of its near edge).
- **Coaching note:** one pick point per foam part. The near edge keeps the foam level over the carton and lines the keyed corner up the same way every time.

Violation: Unit dropped into the carton.

- **Visible cue:** the unit is released above the carton and falls into the cavity, rather than being lowered until its base touches the cavity floor.
- **SOP rule broken:** Step 2 (lower until the base touches down, then release).
- **Coaching note:** carry the unit all the way down into the cavity. Nothing is released above the carton mouth.

Violation: Unit not seated in the cavity.

- **Visible cue:** the unit rests on the rim of the foam, sits across the cavity at an angle, or has one corner riding on the foam instead of sitting down inside on all sides.
- **SOP rule broken:** Step 2 (lower square to the walls until the unit sits down in the cavity on all sides).
- **Coaching note:** line the body up with the cavity before lowering. If it will not seat, lift clear and re-aim rather than pushing it in.

Violation: Manual missing.

- **Visible cue:** the carton is closed with no manual inside, or the manual is left sitting at the manual station.
- **SOP rule broken:** Step 3 (add one manual before the top pad goes in).
- **Coaching note:** check the manual is in the carton before the top foam pad is laid.

Violation: Manual placed outside the walls.

- **Visible cue:** the manual hangs over a carton wall, sits on top of a flap, or stands above the top edge of the walls when the flaps are closed.
- **SOP rule broken:** Step 3 (lay the manual flat inside the carton walls, with nothing above the top edge).
- **Coaching note:** push the manual down inside the walls. Anything over the edge gets crushed by the flaps.

Violation: Top foam pad missing.

- **Visible cue:** the flaps are closed with no top foam pad over the load, or the pad is left on the table at the end of the episode.
- **SOP rule broken:** Step 4 (lay one top foam pad flat over the load before the flaps are closed).

- **Coaching note:** the top pad is a pack layer, not an option. It goes in over the manual every time.

Violation: Top foam pad not flat or outside the walls.

- **Visible cue:** the top pad sits on a carton wall or a flap, stands above the top edge, or has a corner riding up a wall when the flaps are closed over it.
- **SOP rule broken:** Step 4 (lay the pad flat inside the walls, pressed down with nothing above the top edge).
- **Coaching note:** press the pad down inside the walls. A pad on the wall edge stops the flaps closing flat.

Violation: Load left loose (void not closed).

- **Visible cue:** the load shifts or a void is left along a wall when it is pressed after the manual goes in, and the operator lays the top foam pad and carries on instead of re-seating the load.
- **SOP rule broken:** Step 3 (press the load down and re-seat any layer that shifts before the top pad goes in).
- **Coaching note:** press the load before the top pad goes in. A layer that shifts under the press is not seated.

Violation: Carton lifted or tilted while open.

- **Visible cue:** the operator picks up, tilts or turns the open carton at any point before it is sealed, including to shake the load down.
- **SOP rule broken:** Step 1 (the carton stays flat at the center until it is sealed).
- **Coaching note:** bring the parts to the carton, never the carton to the parts. An open carton is never lifted.

Violation: Flaps closed in the wrong order.

- **Visible cue:** a major flap is folded down before both side (minor) flaps are in, or a minor flap is folded over a major flap.
- **SOP rule broken:** Step 5 (both side flaps in first, then the two major flaps down over them).
- **Coaching note:** side flaps first, every time. A minor flap folded on top of a major flap leaves the top unsupported.

Violation: Flap forced over a raised load.

- **Visible cue:** a flap is pressed or pushed down onto a load or a top foam pad that stands above the carton walls, bowing the flap or the carton top.
- **SOP rule broken:** Steps 4 and 5 (never force a flap down over a raised load; push the load down inside the walls first).

- **Coaching note:** if a flap will not lie flat, the load is too high. Open up and re-seat rather than forcing the flap.

Violation: Flap not folded in, or flaps overlapped.

- **Visible cue:** a flap is left standing, folded only part way in, or one major flap rides over the other so the closed top is stepped. A small gap along the seam with both flaps folded flat and meeting is normal and is not a violation.
- **SOP rule broken:** Step 5 (fold each flap all the way in, with neither major flap riding over the other).
- **Coaching note:** fold each flap fully and let the two edges meet. Judge the fold, not the width of the seam line.

Violation: Center strip missing or off the seam.

- **Visible cue:** the first strip is not landed along the center seam, is landed off to one side of it, or is skipped so the seam is only covered by the end strips.
- **SOP rule broken:** Step 6.1 (land the center strip along the center seam).
- **Coaching note:** start the H at the center seam and keep the strip centered on it the whole way.

Violation: End strip missing.

- **Visible cue:** only one end seam is taped, or both end strips are skipped, so the finished pattern is a line or a T rather than an H.
- **SOP rule broken:** Steps 6.2 and 6.3 (one strip across each end seam, parallel to each other).
- **Coaching note:** count the strips before ending the episode: one center, two ends, three in total.

Violation: Tape strip landed one-handed.

- **Visible cue:** a strip is taken and landed with one gripper only, rather than the right gripper taking one side, the left gripper taking the other, and both landing it together.
- **SOP rule broken:** Step 6 (both grippers hold the strip, line it up with the seam and land it together).
- **Coaching note:** two grippers on every strip. A one-handed strip folds back on itself before it lands.

Violation: Tape strip does not wrap the wall.

- **Visible cue:** a strip ends on the top face of the carton instead of carrying over the top edge and down the wall, or one wing is much shorter than the other.
- **SOP rule broken:** Step 6 (carry each strip over the top edge and down the wall by about the same amount at both ends).

- **Coaching note:** run each strip past the edge and onto the wall. A strip that stops on the top face peels first.

Violation: Extra tape landed.

- **Visible cue:** more than three strips are landed on the carton top, or a strip is landed twice over the same seam.
- **SOP rule broken:** Step 6 (three strips seal the carton; do not land extra strips).
- **Coaching note:** the H is three strips. Extra tape is waste and hides a seam that was not closed properly.

Violation: Table not cleared.

- **Visible cue:** the episode ends with a foam insert, top foam pad, manual or tape offcut left on the table.
- **SOP rule broken:** Step 7 (the table holds no foam, manual or tape offcut).
- **Coaching note:** clear every station as part of the episode, not after it.

Violation: Wrong arm used for an action.

- **Visible cue:** any step that specifies the left gripper or the right gripper is performed with the opposite gripper. Pressing a landed tape strip is excluded: either gripper may do that.
- **SOP rule broken:** any step that specifies a gripper, including Steps 1, 2, 3, 4, 5 and 6.
- **Coaching note:** operator confusion about left vs right roles. Walk through the SOP step by step with the operator.

Violation: Arms not fully at home position at episode end.

- **Visible cue:** in the final frame, both arms are close to the home position but not exactly at it (gripper not fully open, or arm position visibly off home).
- **SOP rule broken:** Step 7 (return to home position with grippers open).
- **Coaching note:** complete the home motion explicitly before ending recording.

Non-violation failures

These are episode failures that are not the operator's fault and do not go in the violation set. They are recorded as system issues, the episode is discarded/deleted, and the episode does not become a coaching point for the operator.

- **Recording stopped or paused mid-episode:** Cause: software or hardware issue with the recording system.
- **Camera dropped frames or lost feed during the episode:** Cause: camera or capture system issue.
- **Hardware fault on the robot arm:** Cause: gripper malfunction, arm position drift, or motor error during the episode.

- **Tape strip lost its tack or tore:** Cause: a pre-cut strip supplied dried out, curled on itself or tore in handling, not an operator error. Re-stage the tape station before the next episode.
- **Carton arrived damaged:** Cause: a crushed wall, torn flap or open bottom seal on the carton as supplied. Replace the carton before the next episode.
- **Foam arrived the wrong size:** Cause: a bottom insert whose cavity does not match the unit model, or a top pad that does not fit inside the carton walls, through no fault of the operator. Replace the foam before the next episode.
- **Unit arrived damaged:** Cause: a dented unit, or one with a torn bag, supplied to staging. Remove it before the next episode.

After each episode: repeat or reset

- **Batch sessions (e.g., 4x):** if fewer than the target number of units are packed, reset the workspace and return to Step 1 with the next carton. Once the target number is packed, reset the workspace before the next session.
- Clear the table and re-run the Setup checklist before the next session.

After the episode: reset the workspace

This part is not recorded. It is just how you reset the table for the next episode.

- With recording off, peel the three tape strips and both wings off the sealed carton and bin them.
- Open the four flaps and stand them upright again.
- Lift out the top foam pad and return it to the left-side foam source.
- Lift out the manual and return it to the pile at the front-center manual station.
- Lift the unit out of the cavity and return it, in its bag, to the start zone for the next episode's config — front-left (Config L), back-center (Config M) or front-right (Config R).
- Lift the bottom foam insert out of the carton and return it to the left-side foam source, cavity side up.
- Square the empty carton at the center carton station, flaps upright, bottom seal intact.
- Lay out three fresh pre-cut tape strips at the center-right tape station, tacked by one edge.
- Wipe the table and confirm it holds nothing else.

Warning: The table must be clear except for the unit in its start zone, the carton at the center, the foam source at the left side, the manual station at the front-center, and the pre-cut tape strips at the center-right.

Annotation subtasks

1. Seat the bottom foam insert with the left gripper (pinched by its near edge, cavity up, keyed corner matched, pressed flat)

2. Place the unit in the cavity (lowered until it touches down)
3. Add the manual flat over the unit and foam
4. Lay the top foam pad flat over the load, inside the carton walls (grabbed at the middle of its near edge)
5. Close the flaps (both side flaps in, then both major flaps down)
6. Land the center strip along the center seam with both grippers
7. Land the right end strip
8. Land the left end strip to complete the H-pattern
9. Confirm the pack-out and home the arms